SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1153

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

QUESTIONS: 05 **Total Marks:35**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I **K Madan Mohan Reddy (192411206)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Byreddy Madan

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q2. Scenario: Food Delivery Application System
Features:
Browse Menu, Place Order, Payment, Order Tracking

Usability Constraints:

- Order placement within 4 clicks
- Real-time tracking visibility
- Mobile-friendly interface

Question:
Apply **usability testing techniques** to evaluate the system. Identify usability issues and propose improvements based on the constraints. Explain how these changes enhance user experience. *(BL4)*

Usability Testing for Food Delivery Application System

Introduction

The Food Delivery Application System allows users to browse menus, place food orders, make payments, and track deliveries. For the application to be successful, it must provide a smooth, efficient, and user-friendly experience. The given usability constraints are that users should be able to place an order within four clicks, view real-time order tracking information, and access the application easily on mobile devices. Usability testing is therefore essential to evaluate whether the system meets these requirements and to identify any problems that may affect user satisfaction and performance.

Usability Testing Techniques

Various usability testing techniques can be used to evaluate the system. Task-based testing is one of the most effective methods, where users are asked to perform common tasks such as searching for a food item, adding it to the cart, completing payment, and tracking the order. The time taken, number of clicks, and errors made by users are recorded. Think-aloud testing is another technique in which users verbalize their thoughts while interacting with the application. This helps testers understand user expectations and identify confusing elements in the interface.

- **Task-Based Testing:** Observe users while browsing menus, ordering food, making payments, and tracking orders.
- **Think-Aloud Testing:** Users explain their thoughts while performing tasks.
- **Heuristic Evaluation:** Experts evaluate the interface using usability principles.
- **Mobile Usability Testing:** Test the application on different mobile devices and screen sizes.
- **User Satisfaction Survey:** Collect feedback regarding ease of use and overall experience.

Usability Metrics Measured

- Task completion rate
- Number of clicks to place an order
- Time taken to complete an order
- Number of user errors
- User satisfaction score
- Tracking accuracy and visibility
- Mobile responsiveness

Identified Usability Issues

The usability testing process may reveal several issues. Users may find that placing an order requires navigating through multiple screens, resulting in more than four clicks. Some users may struggle to locate food items because menu categories are not organized properly. The checkout process may contain too many form fields, cause delays and increasing user effort.

Another common issue is the lack of clear and timely order tracking updates. Users may not receive accurate information regarding order preparation, dispatch, or estimated delivery time. On mobile devices, small buttons, crowded layouts, and unreadable text may create interaction difficulties. Some users may also experience slow page loading or accidental clicks due to poor interface design.

- More than 4 clicks required to place an order.
- Confusing menu structure and navigation.

- Slow checkout and payment process.
- Tracking information not updated in real time.
- Small buttons and unreadable text on mobile devices.
- Lack of clear feedback after actions.
- Difficulty locating important features.

Proposed Improvements

Simplifying the Ordering Process

The application should provide a streamlined ordering workflow. Features such as "Quick Order," "Repeat Previous Order," and saved customer preferences can reduce the number of clicks required. Essential actions such as adding items to the cart and proceeding to checkout should be clearly visible on every screen.

Ordering Process

- Introduce one-click reorder.
- Save user addresses and preferences.
- Provide quick checkout options.
- Reduce unnecessary screens.

Improving Menu Navigation

Food items should be organized into meaningful categories such as Pizza, Burgers, Beverages, and Desserts. Search bars, filters, and sorting options can help users find their desired items quickly. Popular and recommended items should be highlighted to improve discoverability.

- Categorize food items clearly.
- Add search functionality.
- Include filtering and sorting options.
- Display popular and recommended items.

Enhancing the Payment Process

The payment process should be simplified by reducing unnecessary data entry. Secure payment methods such as UPI, credit cards, debit cards, and digital wallets should be integrated. Saving payment details securely for future purchases can further reduce checkout time.

- Support UPI, wallets, and cards.
- Save payment methods securely.

- Minimize form filling.

Providing Better Real-Time Tracking

A GPS-enabled tracking system should display the current location of the delivery partner on a map. Status updates such as "Order Confirmed," "Preparing Food," "Out for Delivery," and "Delivered" should be displayed clearly. Push notifications can keep users informed about important delivery milestones.

Order Tracking

- Implement GPS-based live tracking.
- Show delivery partner location.
- Display estimated delivery time.
- Send real-time notifications.

Optimizing Mobile Usability

The application should use responsive design techniques to adapt automatically to different screen sizes. Buttons should be larger and touch-friendly, while text should remain clear and readable. Important features should be accessible without excessive scrolling, and page loading speed should be optimized.

Mobile Interface

- Use responsive design.
- Increase button size.
- Improve text readability.
- Optimize touch interactions.

Adding Feedback and Error Handling

Users should receive immediate feedback when actions are performed successfully. For example, confirmation messages should appear after adding items to the cart or completing payments. Clear error messages should guide users in correcting mistakes without causing frustration.

Benefits of Improvements

- Faster order placement.
- Reduced user effort.
- Better navigation experience.
- Increased customer satisfaction.
- Higher trust through real-time tracking.
- Improved accessibility on mobile devices.
- Fewer user errors.
- Better overall user experience.

How These Changes Enhance User Experience

The proposed improvements directly enhance user experience by making the application more efficient, effective, and satisfying. Reducing the number of clicks speeds up order placement and minimizes user effort. Improved menu organization allows users to find products quickly, saving time and reducing frustration. Simplified payment options increase convenience and reduce cart abandonment rates.

Real-time tracking enhances transparency and builds trust because users can monitor their orders at every stage. Mobile-friendly design ensures accessibility for a larger audience and provides a smooth experience across different devices. Better feedback mechanisms improve user confidence by clearly communicating system status and preventing confusion.

Expected Outcomes

- Orders completed within 4 clicks.
- Improved task completion rate.
- Reduced ordering time.
- Enhanced customer engagement.
- Increased customer retention.
- Better application usability and efficiency.

System Features:

- Browse Menu
- Place Order
- Payment
- Order Tracking

Usability Constraints:

- Order placement within 4 clicks
- Real-time tracking visibility
- Mobile-friendly interface

Conclusion

Usability testing techniques such as task-based testing, think-aloud testing, mobile testing, heuristic evaluation, and user surveys are effective in evaluating the Food Delivery Application. The identified usability issues can be resolved through interface simplification, improved navigation, streamlined payment methods, enhanced tracking features, and responsive design. These changes help meet the usability constraints and provide a more efficient, user-friendly, and satisfying experience for customers.

The evaluation revealed issues such as excessive clicks during order placement, complicated menu navigation, lengthy payment procedures, insufficient real-time tracking visibility, and mobile interface challenges. Addressing these issues through simplified workflows, improved navigation, faster payment options, live GPS tracking, and responsive mobile design significantly enhances the application's usability.

The proposed improvements help ensure that users can place orders quickly within the required four clicks, track their orders accurately in real time, and access all features conveniently on mobile devices. These enhancements reduce user effort, minimize errors, improve efficiency, and increase overall satisfaction. A well-designed and user-friendly interface encourages users to complete transactions more frequently and builds trust in the service.